

TWITTER ANALYSIS – USING POWERBI

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1. Introduction

This report summarizes the development and execution of nine analytical tasks using Power BI, focusing on Twitter data. The project aimed to apply specific business rules, DAX calculations, and dynamic time-based visualizations to build interactive, conditionally-filtered dashboards.

2. Background

The project involved creating advanced Power BI visuals to analyze tweet engagement, impressions, and interaction types. Each task required strict filtering based on tweet metrics (likes, retweets, replies, impressions, word/character count, posting time) and dashboard visibility control depending on the system time (IST).

The goal was to demonstrate expertise in dynamic filtering, DAX measure design, data transformation, and business rule implementation.

3. Learning Objectives

- Understand advanced DAX to filter and conditionally display data.
 - Apply dynamic time-based visibility for visuals (using IST).
 - Handle multiple data validation conditions (even/odd checks, character limits, specific letter exclusions).
 - Manage blank data scenarios professionally in Power BI.
 - Create clean, user-friendly interactive dashboards.
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4. Activities and Tasks

The nine tasks completed were:

Task Description

1. Develop a chart that displays tweets with the highest engagement rates (top 10%). Include only those tweets that have received more than 50 likes and were posted on weekdays and this graph should work only between 3PM IST to 5 PM IST apart from that time we should not show this graph in dashboard itself as well as tweet character count should be below 30.

2. Create a visual that shows the average engagement rate and total impressions for tweets posted between '01-01-2020' and '30-06-2020'. Filter out tweets that received fewer than 100 impressions and like should be 0 and this graph should work only between 3PM IST to 5 PM IST apart from that time we should not show this graph in dashboard itself.
3. Build a chart to identify the top 10 tweets by the sum of retweets and likes. Filter out tweets posted on weekends and show the user profile that posted each tweet and this graph should work only between 3PM IST to 5 PM IST apart from that time we should not show this graph in dashboard itself and the tweet impression should be even number and tweet date should be odd number as well as tweet word count be below 30
4. Create a clustered bar chart that breaks down the sum of URL clicks, user profile clicks, and hashtag clicks by tweet category (e.g., tweets with media, tweets with links, tweets with hashtags). Only include tweets that have at least one of these interaction types and this graph should work only between 3PM IST to 5 PM IST apart from that time we should not show this graph in dashboard itself and the tweet date should be even number as well as tweet word count be above 40.
5. Plot a scatter chart to analyse the relationship between media engagements and media views for tweets that received more than 10 replies. Highlight tweets with an engagement rate above 5% and this graph should work only between 6PM IST to 11 PM IST apart from that time we should not show this graph in dashboard itself and the tweet date should be odd number as well as tweet word count be above 50.
6. Create a line chart showing the trend of the average engagement rate over each month of the year. Separate the lines for tweets with media content and those without and this graph should work only between 3PM IST to 5 PM IST and 7 AM to 11AM apart from that time we should not show this graph in dashboard itself and the tweet engagement should be even number and tweet date should be odd number as well as tweet character count should be above 20 and need to remove tweet word which has letter 'C'.
7. Create a dual-axis chart that shows the number of media views and media engagements by the day of the week for the last quarter. Highlight days with significant spikes in media interactions. this graph should work only between 3PM IST to 5 PM IST and 7 AM to 11AM apart from that time we should not show this graph in dashboard itself and the tweet impression should be even number and tweet date should be odd number as well as tweet character count should be above 30 and need to remove tweet word which has letter 'H'.

8. Analyse tweets to show a comparison of the engagement rate for tweets with app opens versus tweets without app opens. Include only tweets posted between 9 AM and 5 PM on weekdays and this graph should work only between 12PM IST to 6PM IST and 7 AM to 11AM apart from that time we should not show this graph in dashboard itself and the tweet impression should be even number and tweet date should be odd number as well as tweet character count should be above 30 and need to remove tweet word which has letter 'D'.

9. Develop a visualization that compares the number of replies, retweets, and likes for tweets that have received media engagements greater than the median value. Include a filter for tweets posted in between June and August of 2020 and this graph should work only between 3PM IST to 5 PM IST and 7 AM to 11AM apart from that time we should not show this graph in dashboard itself and tweet date should be odd number and media views should be even number as well as tweet character count should be above 20 and need to remove tweet word which has letter 'S'.

5. Skills and Competencies

- Advanced DAX formula writing (PERCENTILEX.INC, SWITCH, IF, CALCULATE).
- Dynamic time-based chart visibility using NOW() and HOUR().
- Data cleansing (handling character count, word count, and exclusions).
- Visual customization (combo charts, scatter, line, clustered column).
- Handling blank datasets using fallback text messaging techniques.
- Cross-validation of data (even/odd numbers, letter exclusion).
- Error troubleshooting and debugging of DAX errors.

6. Feedback and Evidence

- Successfully implemented dynamic dashboard behavior based on system time (IST).
 - Clean handling of blank charts where no data matched strict conditions.
 - Effective application of complex business rules to filter the dataset.
 - Feedback included identifying situations where conditions were too strict (e.g., no tweets without 'S' or with word count above 50) and creating fallback solutions.
 - Dashboards maintained clean and user-friendly appearance even when visuals were empty.
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7. Challenges and Solutions

Challenge	Solution
DAX errors using measures in filter pane	Shifted to dynamic visibility control using conditional formatting instead.
No data matching strict filter conditions (e.g., character count >40)	Implemented fallback message cards dynamically using DAX.
Date parsing and quarter filtering errors	Corrected parsing logic, ensured timezone consistency (UTC/IST).
Controlling chart visibility for charts (not just cards)	Used BLANK() strategy inside DAX measures instead of trying to hide visuals.
Heavy data conditions causing blank visuals	Communicated properly to users with friendly text explanations.

8. Outcomes and Impact

- Created fully functional, business-rule-driven Power BI dashboards.
 - Mastered advanced DAX and time-based visibility controls.
 - Improved handling of special cases when no data matched the conditions.
 - Gained confidence to handle dynamic filters, complex visuals, and real-world dashboard requirements.
 - Developed clean, dynamic dashboards ready for real-time business reporting under strict data conditions.
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9. Conclusion

This Power BI project successfully demonstrated the application of complex data filtering, dynamic time-based visual control, and advanced DAX techniques to create professional dashboards.

Despite challenging data conditions and strict requirements, effective solutions were implemented to ensure the dashboards were interactive, clean, and business-ready.

The project enhanced technical skills significantly and established a strong foundation for solving complex Power BI reporting needs in the future.

